(c) The inhaled CO adversely affects the respiratory centre in the brain.
(d) the inhaled CO adversely affects the respiratory centre in the brain.

Ans. (b) : Carbon Monoxide pollution occurs primarily from emissions produced by fossil fuel powered engines. The incomplete reaction of air with fuel produces the colourless, odourless and highly toxic gas. The main issue with Carbon Monoxide is its health effects. It is capable of binding to the chemicals in our blood, called haemoglobin. It does so far more effectively than oxygen and also stays bound to the haemoglobin for far longer than oxygen does. The effect is that the blood is starved of oxygen, which then affects the rest of the body.

70. Indiscriminate disposal of used florescent electric lamps causes mercury pollution in the environment. Why is mercury used in the manufacture of these lamps?
(a) A mercury coating on the inside of the lamp makes the light bright white
(b) When the lamp is switched on, the mercury in the lamp causes the emission of ultra-violet radiations
(c) When the lamp is switched on, it is the mercury which converts the ultra-violet energy into visible light.
(d) None of the statement given above is correct about the use of mercury in the manufacture of fluorescent lamps

Ans. (c) : When the lamp is switched on, it is the mercury which converts the ultra-violet energy into visible light. .

71. Hydrogen fuel cell vehicles produce one of the following as "exhaust"
(a) NH_3 (b) CH_4
(c) H_2O (d) H_2O_2

Ans. (c) : A hydrogen vehicle is an alternative fuel vehicle that uses hydrogen as its onboard fuel for motive power. The hydrogen vehicle use hydrogen fuel cell for generation of motive power. These fuel cell in which hydrogen serve as a fuel and oxygen as an oxidant emits exhaust of water.

72. Consider the following:
1. Oxides of Hydrogen
2. Oxides of Nitrogen
3. Oxides of Sulphur
Which of the above causes/cause acid rain?
(a) 1 and 2 only (b) 3 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (c) : Acid rain is caused by Oxides of Nitrogen, Oxides of Sulphur.

73. If a potato is placed on a pure paper plate which is white and unprinted and put in a microwave oven, the potato heats up but the paper plate does not. This is because:
(a) Potato is mainly made up of starch whereas paper is mainly made up of cellulose
(b) Patato transmits microwaves whereas paper reflects microwaves
(c) Potato contains water whereas paper does not contain water
(d) Potato is a fresh organic material whereas paper is a dead organic material

Ans. (c) : Potato contains water whereas paper does not contain water.

74. Chlorination is a process used for water purification. The disinfecting action of chlorine is mainly due to
(a) The formation of hydrochloric acid when chlorine is added to water
(b) The formation of hypochlorous acid when chlorine is added to water
(c) The formation of nascent oxygen when chlorine is added to water
(d) The formation of hydrogen when chlorine is added to water

Ans. (b) : Water chlorination is the process of adding chlorine or hypochlorite to water this method is used to kill certain bacteria and other microbes as Chlorine is highly toxic.

BIOLOGY

75. Widespread resistance of malarial parasite to drugs like chloroquine has prompted attempts to develop a malarial vaccine to combat malaria. Why is it difficult to develop an effective malaria vaccine?
(a) Malaria is caused by several species of plasmodium
(b) Man does not develop immunity to malaria during natural infection
(c) Vaccines can be developed only against bacteria
(d) Man is only an intermediate host and not the definitive host

Ans. (b) : It is difficult to develop an effective malaria vaccine, because repeated natural infection does not produce detectable resistance to exo-erythrocytic cycle of malaria in man.

76. Which one of the following processes in the bodies of living organisms is a digestive process?
(a) Breakdown of proteins into amino acids
(b) Breakdown of glucose into CO_2 and H_2O.
(c) Conversion of glucose into glycogen
(d) Conversion of amino acids into proteins.

Ans. (a) : Dietary protein is degraded into the amino acids and their derivatives are then absorbed by the gastrointestinal tract. Digestion is a catabolic process in which large molecules of the food gets broken out into small substances.

- It typically begins in the stomach when pepsinogen is converted to pepsin by the action of hydrochloric acid and continued by trypsin and chymotrypsin in the intestine. During respiration breakdown of glucose into CO_2 and H_2O takes place.
- Conversion of glucose into glycogen is a glycogenesis process. It occurs in liver.
- During translation, amino acids translated into proteins.

77. From the point of view of evolution of living organisms, which one of the following it the correct sequence of evolution?
(a) Otter-Tortoise-Shark
(b) Shark-Tortoise-Otter
(c) Tortoise-shark-Otter
(d) Shark-Otter-tortoise

Ans. (b) : Correct sequence of evolution:
Shark-Tortoise-Otter.

78. Consider the following statements:
1. Hepatitis B is several times more infectious than HIV/AIDS
2. Hepatitis B can cause liver cancer
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Both 1 and 2 . So, the correct answer is option (c).
Hepatitis B is much more infectious than AIDS. It can cause liver cancer.

79. Consider the following statements:
1. Every individual in the population is equally susceptible host for Swine Flu.
2. Antibiotics have no role in the primary treatment of Swine Flu.
3. to prevent the future spread of Swine Flue in the epidemic area, the swine (pigs) must all be culled.
Which of the statements given above is/are correct?
(a) 1 and 2 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (a) : Swine influenza is a respiratory disease of pigs caused by type A influenza viruses that regularly cause outbreaks of influenza in pigs. Influenza viruses that commonly circulate in swine are called 'swine influenza viruses or swine flu viruses. H1N1 antiviral agents used for treatment and prevention of swine flu are oseltamiuir (Tamiflu) and zanamivir (Relenza). Swine flu is resistant to amantadine (Symmetrel) and rimantadine (Flumadine).

80. With regard to the transmission of the Human Immunodeficiency Virus, which one of the following statements is not correct?
(a) The chances of transmission from female to male are twice as likely as from male to female.
(b) The chances of transmission are more if a person suffers from other sexually transmitted infections
(c) An infected mother can transmit the infection to her baby during pregnancy, at childbirth and by breast feeding.
(d) The risk of contracting infection from transfusion of infected blood is much higher than an exposure to contaminated needle.

Ans. (a) : Human Immunodeficiency Virus (HIV) is the retrovirus that causes Acquired Immunodeficiency Deficiency Syndrome (AIDS). The virus weakens a person's ability to fight infections and cancer by infecting and killing CD4 cells (also called T cells). Women account for more than half the number of people living with HIV worldwide. Young Women (10-24 years old) are twice as likely to acquire HIV as young men the same age. HIV is transmitted by
(i) Infected blood
(ii) Unsterilised injections
(iii) Unprotected sex
(iv) Infants from infected mothers

81. King Cobra is the only snake that makes its own nest. Why does it make its nest?
(a) It is a snake-eater and the nest helps attract other snakes
(b) It is a viviparous snake and needs a nest to give birth to its offspring
(c) It is an oviparous snake and lays its eggs in the nest and guards the next until they are hatched
(d) It is a large, cold blooded animal and needs a nest to hibernate in the cold season

Ans. (c) : It is an oviparous snake and lays its eggs in the nest and guards the nest until they are hatched.

ENVIRONMENT AND ECOLOGY

82. What are the possible limitations of India in mitigating the global warming at present and in the immediate future?
1. Appropriate alternate technologies are not sufficiently available.
2. India cannot invest huge funds in research and development.
3. Many developed countries have already set up their polluting industries in India.
Which of the statements given above is/are correct?
(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3